



Fullstack Web Development

Session 8 | Week 9/Day 1 | 120 min

Laravel - Part03

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Laravel Routing

- **What is Routing?**
 - Routing in Laravel is used to connect a URL with a specific action in the application.
 - A route connects:
 - URL (URI)
 - HTTP Request Method
 - Controller Method / Function
 - Response or View
 - When a user visits a URL in the browser, Laravel checks the route definition and decides what response should be returned.
 - Routes are one of the most important parts of a Laravel application because they control how users access pages and data.
 - When creating a new page in Laravel, you should think about:

Laravel Routing

- When creating a new page in Laravel, you should think about:
 - What URL the user will visit
 - Which controller method will handle the request
 - Which view will be displayed to the user

Laravel Routing

- **Route Files in Laravel**
 - Laravel routes are stored inside the routes folder.
 - **Main Route Files**

File	Purpose
routes/web.php	Handles web application routes
routes/api.php	Handles REST API routes
routes/console.php	Handles console command routes
routes/channels.php	Broadcasting channels

Laravel Routing

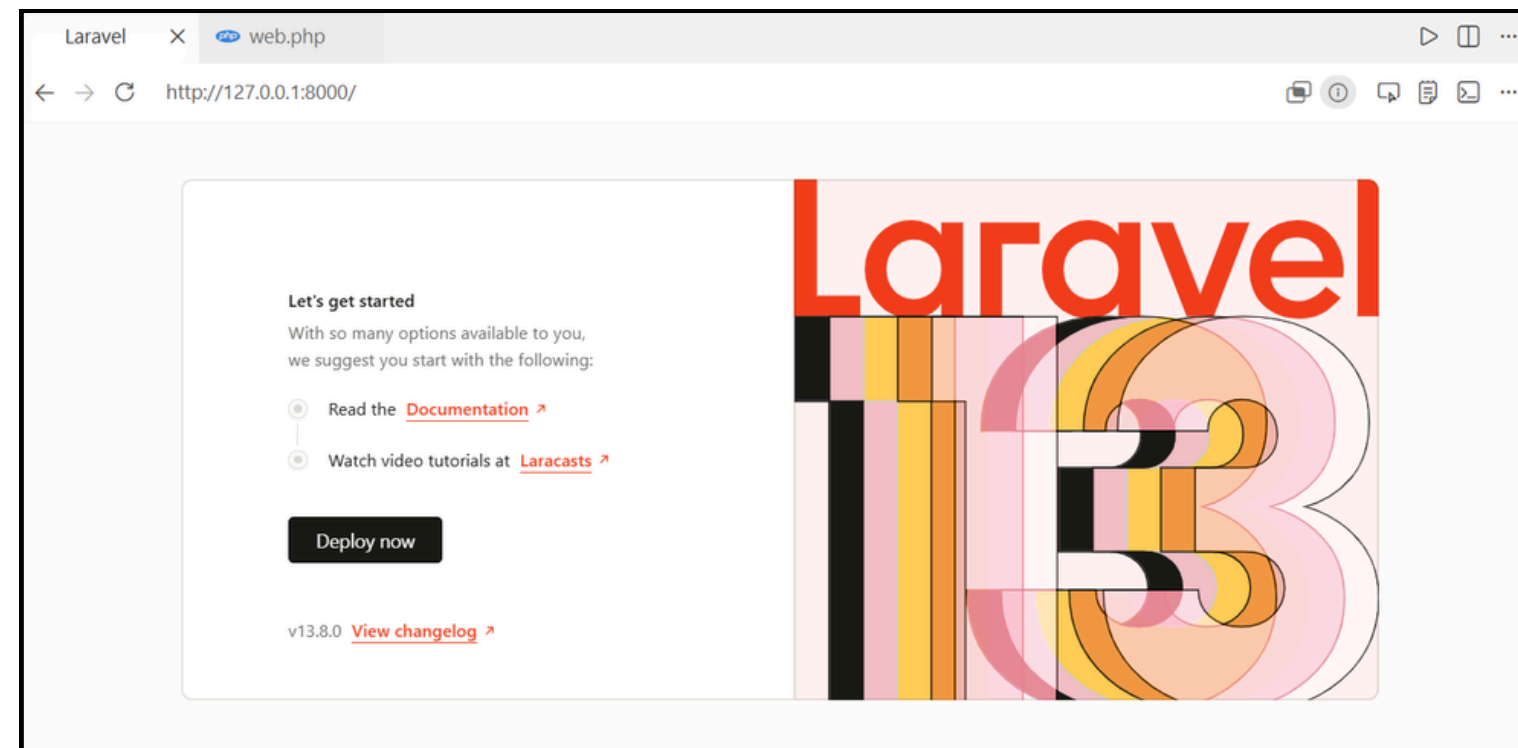
- Web Routes vs API Routes

- Web Routes

- File Location: `routes/web.php`
 - Used for websites with HTML pages
 - Supports sessions, cookies, authentication, and Blade templates

- Example:

```
Route::get('/', function () {  
    return view('welcome');  
});
```



Laravel Routing

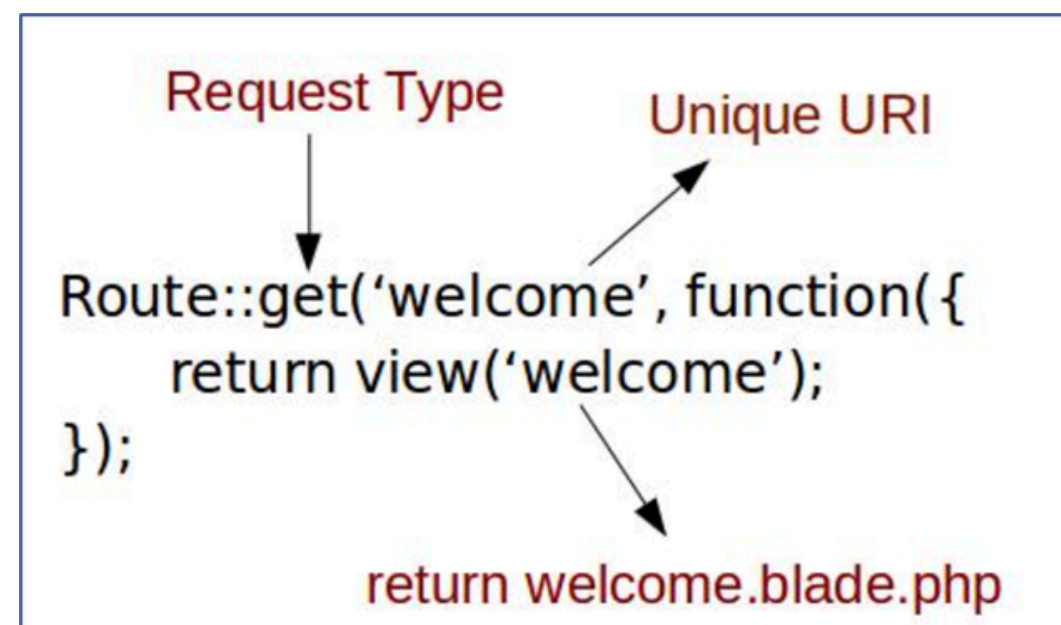
- Web Routes vs API Routes

- Web Routes

- Basic Route Syntax

```
Route::method('URI', function () {  
    return response;  
});
```

- Example:



Laravel Routing

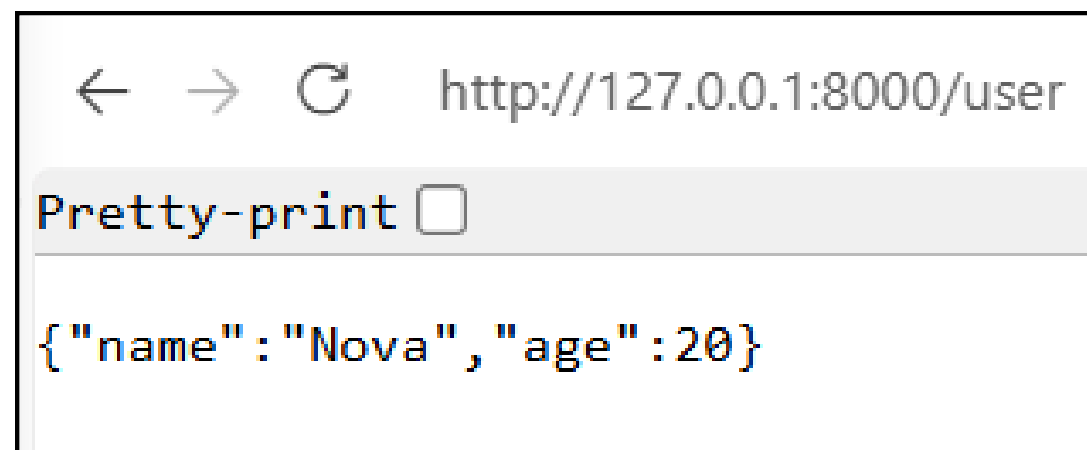
- Web Routes vs API Routes

- API Routes

- File Location: `routes/api.php`
 - Used for mobile apps or frontend frameworks (React/Vue)
 - Usually returns JSON data
 - URL automatically prefixed with `/api`

- Example:

```
Route::get('/user', function () {  
    return response()->json([  
        'name' => 'Nova',  
        'age' => 20  
    ]);  
});
```



```
< > ↻ http://127.0.0.1:8000/user  
Pretty-print ☐  
{"name": "Nova", "age": 20}
```



Laravel Routing

- **What is REST API?**
 - REST API is a standard way to perform CRUD operations using HTTP methods.

CRUD	HTTP Method	Purpose
Create	POST	Create new record
Read	GET	Get/list data
Update	PUT	Replace full record
Update	PATCH	Update part of record
Delete	DELETE	Delete record

Laravel Routing

- **Common HTTP Response Codes**

Code	Meaning
200	OK
201	Created
204	No Content
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
405	Method Not Allowed
409	Conflict
500	Internal Server Error

Laravel Routing

- What is REST API?
 - REST API is a standard way to perform CRUD operations using HTTP methods.

- Example

```
Route::get('/contacts', function () {  
    return 'All contacts';  
});  
  
Route::post('/contacts', function () {  
    return 'Create contact';  
});
```

```
Route::put('/contacts/{id}', function ($id) {  
    return 'Update contact ' . $id;  
});  
  
Route::delete('/contacts/{id}', function ($id) {  
    return 'Delete contact ' . $id;  
});
```

Laravel Routing

- **Simple Routes Without Controller**
 - Laravel allows creating routes directly without using a controller.
 - These routes can return:
 - Text
 - JSON
 - HTML
 - Blade View

Laravel Routing

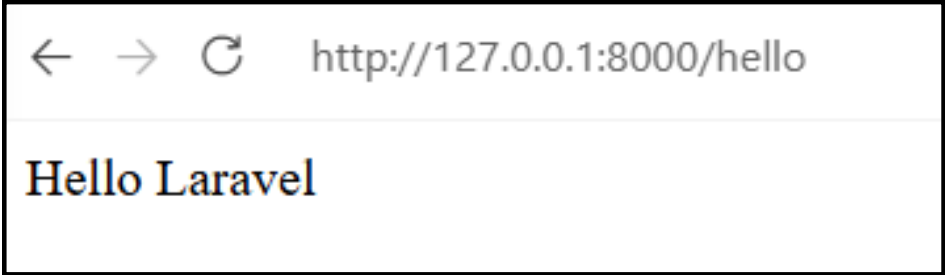
- **Simple Routes Without Controller**

- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return Text**

```
Route::get('/hello', function () {  
    return "Hello Laravel";  
});
```



← → ↻ http://127.0.0.1:8000/hello
Hello Laravel

Part	Description
Route::get()	Handle GET request
/	URL path
function()	Action to execute

Laravel Routing

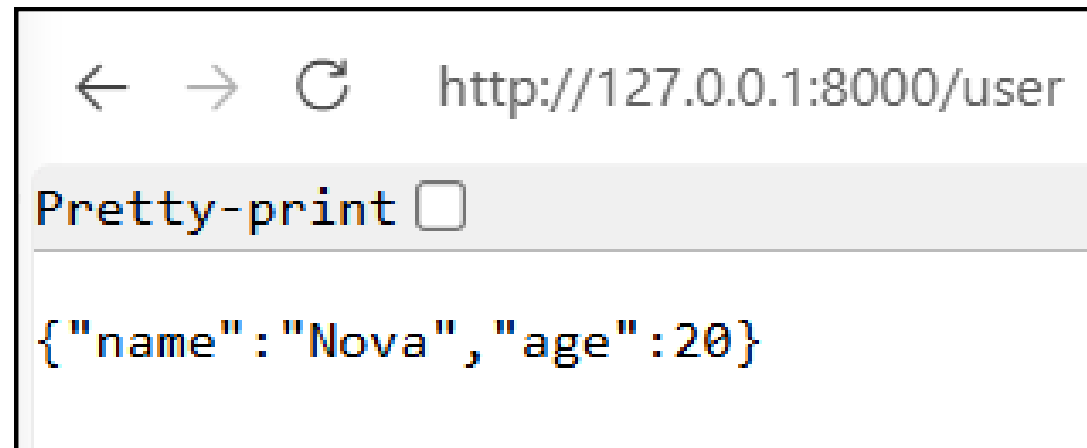
- **Simple Routes Without Controller**

- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return JSON**

```
Route::get('/user', function () {  
    return response()->json([  
        'name' => 'Nova',  
        'age' => 20  
    ]);  
});
```



Laravel Routing

- **Simple Routes Without Controller**

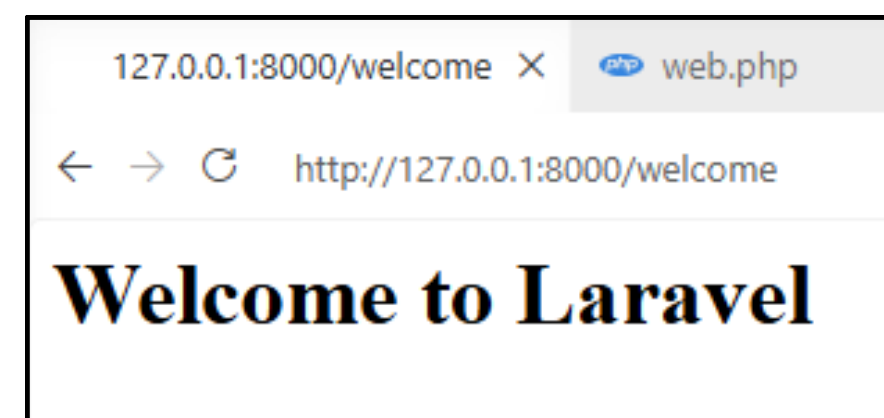
- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return HTML**

```
Route::get('/welcome', function () {  
    return "<h1>Welcome to Laravel</h1>";  
});
```

- Simple HTML



```
Route::get('/about', function () {  
    return "  
        <h1>About Us</h1>  
        <p>This is Laravel Routing Lesson.</p>  
    ";  
});
```

- Return Multiple HTML Elements



Laravel Routing

- **Simple Routes Without Controller**

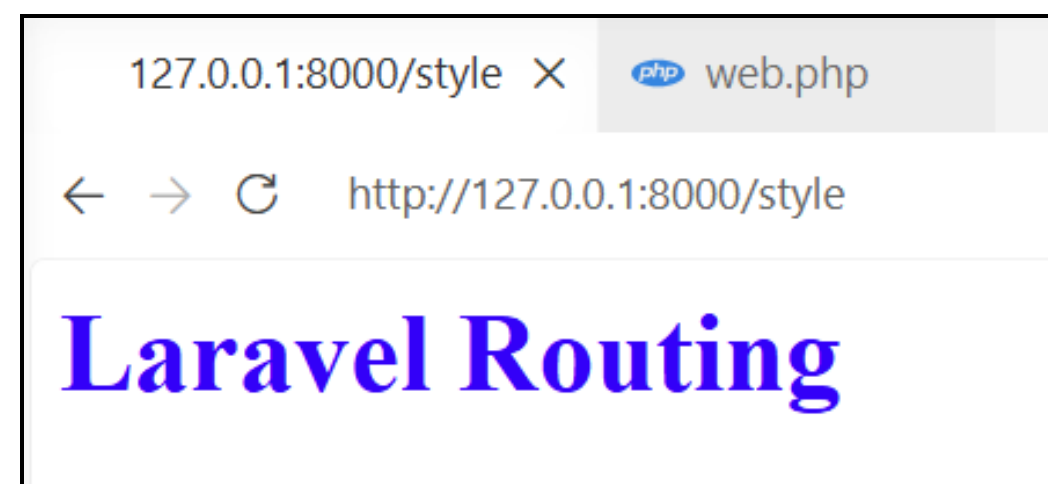
- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return HTML**

```
Route::get('/style', function () {  
    return "  
        <h1 style='color:blue'>  
            Laravel Routing  
        </h1>  
    ";  
});
```

- HTML with Inline CSS



Laravel Routing

- **Simple Routes Without Controller**

- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return HTML**

- Suitable For
 - Testing routes
 - Small examples
 - Learning purposes
 - Quick debugging
- Not Recommended For
 - Large web pages
 - Complex UI designs

- For large applications, it is better to use:

- Controllers
- Blade Templates
- Components

Laravel Routing

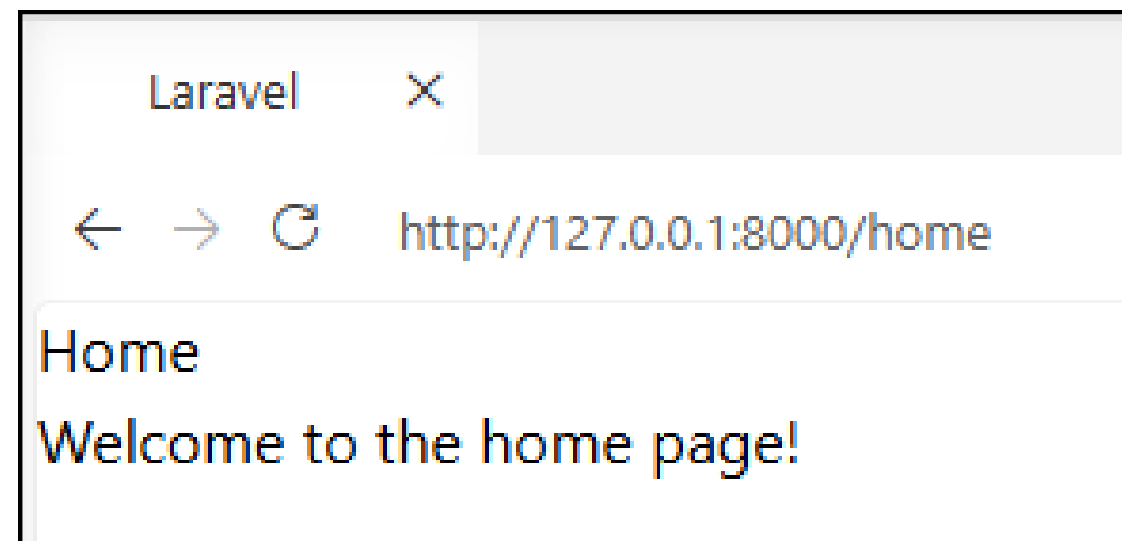
- **Simple Routes Without Controller**

- Laravel allows creating routes directly without using a controller.

- These routes can return:

- **Return View**

```
Route::get('/home', function () {  
    return view('home');  
});
```



- This will return: `resources/views/home.blade.php`

Laravel Routing

- **Simple Routes Without Controller**

- Laravel allows creating routes directly without using a controller.

- Comparison

Method	Best Use
Return Text	Simple messages
Return HTML	Quick HTML output
Return JSON	API response
Return View	Real web pages

Laravel Routing

- Route with Controller
 - Using Controller Method

- Example:

```
use App\Http\Controllers\ContactController;  
  
Route::get('/contacts', [ContactController::class, 'index']);
```



ID	Name	Phone
1	Nova	012345678
2	Dara	098765432
3	Sokha	011223344

Laravel Routing

- **Checking Existing Routes**

- Laravel Artisan command: `php artisan route:list`
- Other useful commands:
 - `php artisan route:list --except-vendor`
 - `php artisan route:list --only-vendor`
 - `php artisan route:list --path=contacts`
 - `php artisan route:list -h`

Laravel Routing

- **Routes for Multiple HTTP Verbs**

- `match()` – Registers the route to respond to get and post method

```
Route::match(['get', 'post'], '/contact', function () {  
    return "Hello World";  
});
```

- `any()` – Registers the route to respond to any HTTP method

```
Route::match(['get', 'post'], '/contact', function () {  
    return "Hello World";  
});
```


Laravel Routing

- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Required Parameter**

- Example:

```
Route::get('/contacts/{id}', function ($id) {  
    return "Contact " . $id;  
});
```

- URL:

- /contacts/1
- /contacts/20



Laravel Routing

- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Multiple Parameters**

- Example:

```
Route::get('/users/{id}/posts/{post}', function ($id, $post) {  
    return "User: $id Post: $post";  
});
```

A screenshot of a web browser. The address bar shows the URL `http://127.0.0.1:8000/users/5/posts/10`. Below the address bar, the page content displays `User: 5 Post: 10`.

← → ↻ `http://127.0.0.1:8000/users/5/posts/10`
`User: 5 Post: 10`

Laravel Routing

- **Route Parameters**
 - Route parameters allow dynamic values in URLs.
- **Optional Route Parameters**
 - Example:

```
Route::get('/companies/{name?}', function ($name = null) {  
    if ($name) {  
        return "Company " . $name;  
    }  
    return "All Companies";  
});
```

← → ↻ http://127.0.0.1:8000/companies

All Companies

← → ↻ http://127.0.0.1:8000/companies/Google

Company Google

Laravel Routing

- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Route Parameter Constraints**

- Example: Only Numbers

```
Route::get('/user/{id}', function ($id) {  
    return $id;  
})->whereNumber('id');
```



← → ↻ http://127.0.0.1:8000/user/100

User ID : 100

Laravel Routing

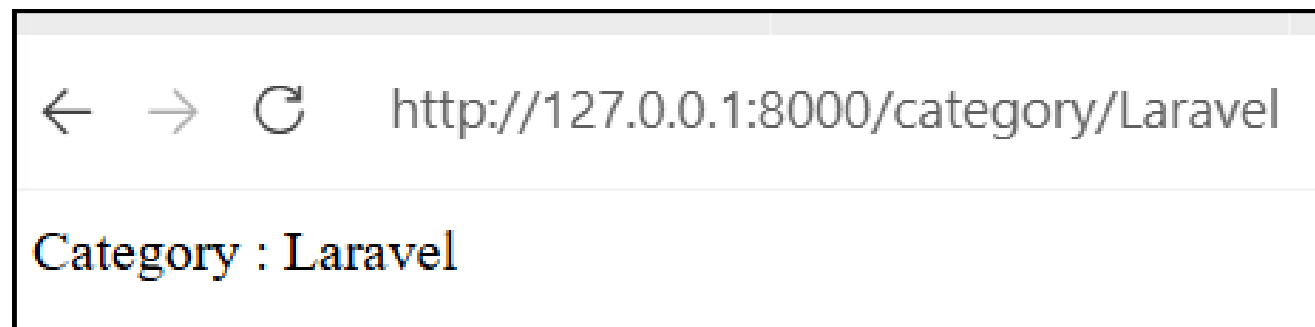
- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Route Parameter Constraints**

- Example: Only Alphabetic Characters

```
Route::get('/category/{name}', function ($name) {  
    return $name;  
})->whereAlpha('name');
```



Laravel Routing

- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Route Parameter Constraints**

- Example: Alpha Numeric

```
Route::get('/product/{code}', function ($code) {  
    return $code;  
})->whereAlphaNumeric('code');
```



← → ↻ http://127.0.0.1:8000/product/A123

Product Code : A123

Laravel Routing

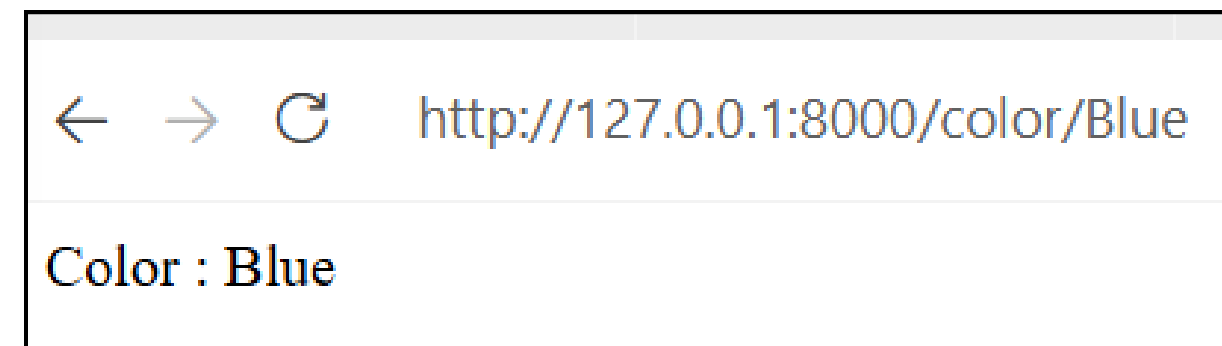
- **Route Parameters**

- Route parameters allow dynamic values in URLs.

- **Route Parameter Constraints**

- Example: Regular Expression Constraint

```
Route::get('/color/{name}', function ($name) {  
    return $name;  
})->where('name', '[A-Za-z]+');
```



Laravel Routing

- **Named Routes**

- Named routes allow developers to generate URLs without hardcoding paths.

- **Creating Named Routes**

```
Route::get('/contacts', function () {  
    return "All Contacts";  
})->name('contacts.index');
```

- **More Named Routes**

```
Route::get('/contacts/{id}', function ($id) {  
    return "Contact " . $id;  
})->name('contacts.show');
```

```
Route::get('/contacts/create', function () {  
    return "Add Contact";  
})->name('contacts.create');
```


Laravel Routing

- **Generating URLs Using `route()`**
 - Named routes provide an easy way to generate URLs or redirects for specific routes without hardcoding the URLs in your code

```
route('contacts.index');  
route('contacts.show', 1);
```



```
Route::get('/home', function () {  
  
    $html = "  
    <h1>Contact App</h1>  
  
    <a href='" . route('contacts.index') . "'>  
        All Contacts  
    </a>  
  
    <a href='" . route('contacts.create') . "'>  
        Add Contact  
    </a>  
  
    <a href='" . route('contacts.show', 1) . "'>  
        Show Contact  
    </a>  
    ";  
  
    return $html;  
});
```

Laravel Routing

- **Route Groups**

- Route groups help organize routes and share attributes.

- **Prefix Route Groups**

```
Route::prefix('admin')->group(function () {  
  
    Route::get('/dashboard', function () {  
        return "Admin Dashboard";  
    });  
  
});
```

- URL: /admin/dashboard

Laravel Routing

- **Route Groups**

- Route groups help organize routes and share attributes.

- **Named Route Groups**

```
Route::prefix('admin')
    ->name('admin.')
    ->group(function () {

        Route::get('/contacts', function () {
            return "All Contacts";
        }->name('contacts.index'));

    });
```

- Generated route name: **admin.contacts.index**

Laravel Routing

- **Route Groups**

- Route groups help organize routes and share attributes.

- **Middleware Route Groups**

```
Route::middleware(['auth'])->group(function () {  
  
    Route::get('/profile', function () {  
        return "User Profile";  
    });  
  
});
```

Laravel Routing

- **Route Groups**

- Route groups help organize routes and share attributes.

- **Controller Route Groups**

```
Route::controller(ContactController::class)->group(function () {  
  
    Route::get('/contacts', 'index');  
    Route::get('/contacts/create', 'create');  
    Route::post('/contacts', 'store');  
  
});
```

Laravel Routing

- **Route Groups**

- Route groups help organize routes and share attributes.

- **Redirect Routes**

```
Route::redirect('/home', '/dashboard');
```

Laravel Routing

- **Fallback Routes**

- Fallback routes execute when no route matches the request.

```
Route::fallback(function () {  
    return "<h1>Sorry, Page Not Found</h1>";  
});
```

- **View Routes**

- Shortcut for returning Blade views.

```
Route::view('/about', 'about');
```

- **Redirect Routes**

```
Route::redirect('/home', '/dashboard');
```

Laravel Routing

- **Route Model Binding**

- Automatically retrieve model data from route parameter.

```
Route::get('/users/{user}', function ($user) {  
    return $user;  
});
```

- Using Model:

```
use App\Models\User;  
  
Route::get('/users/{user}', function (User $user) {  
    return $user;  
});
```


Laravel Routing

- **Resource Routes**

- Laravel provides automatic **CRUD routes**.

```
Route::resource('contacts', ContactController::class);
```

HTTP Method	URI	Controller Method
GET	/contacts	index
GET	/contacts/create	create
POST	/contacts	store
GET	/contacts/{id}	show
GET	/contacts/{id}/edit	edit
PUT/PATCH	/contacts/{id}	update
DELETE	/contacts/{id}	destroy

Laravel Routing

- **Route Middleware**

- Middleware filters HTTP requests.

```
Route::get('/dashboard', function () {  
    return "Dashboard";  
})->middleware('auth');
```

- **CSRF Protection in Routes**

- Laravel automatically protects POST routes using CSRF tokens.

- Blade Example:

```
<form method="POST">  
    @csrf  
</form>
```



THANK YOU

For your attention !

